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Research

Under Attack

CYBERSECURITY | THEMATIC RESEARCH | OCTOBER 2026

AI is rapidly lowering the barrier to cyberattacks on corporates, government and critical infrastructure. This report outlines the growing risk and why it now warrants priority across corporate strategy, investment research, supply-chain resilience and critical infrastructure.

Supported by a dbDataInsights (dbDIG) survey of more than 50 cybersecurity experts

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Source: dbDataInsights survey of cybersecurity experts, September 2026. Word cloud from participant responses.

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Report Introduction

Our proprietary dbDataInsights survey shows cyber threats are evolving faster than organisations can adapt

"It's the only thing that can shut down my business."

That was the punchline from a conversation I had recently with the CEO of a large privately held financial services company. This was his comment when I followed up as to why cybersecurity risk is the number one concern that "keeps him up at night".

To help understand this risk, our dbDataInsights team surveyed more than 50 cybersecurity experts. The uncomfortable conclusion is that many firms are unprepared for cyber threats. Cyber risk is increasing as AI adoption accelerates.

—89% agree the threat landscape is evolving faster than organisations can adapt.

—78% say the gap between highly and poorly prepared organisations is widening.

What has changed with AI advancement? Cyberattacks have become faster, cheaper, more deceptive and more scalable. Capabilities once reserved for sophisticated actors are now broadly accessible to a new generation of profit-driven criminals. At the same time, rising geopolitical tension has made cyber operations an increasingly attractive low-cost, deniable tool for nation states. Corporates and investors therefore need to understand their broader exposure and critical infrastructure risk in their supply chain and with stakeholders.

Among other autonomous incidents, the recent Hugging Face incident highlighted AI's cyberattack potential. AI agents caused harm without direction from a human attacker, with weak guardrails and no human judgement to restrain them. Beyond the systemic questions it raised, the attack highlights three additional concerns: (i) AI operates at a speed and scale humans cannot match, with agents taking over 17,000+ actions in four days; (ii) the intrusion began in the data-processing pipeline, where AI platforms are uniquely exposed; and (iii) the agents escaped their test environment through a widely used third-party tool representing supplier risk in an unfamiliar form.

89%

agree the cyber threat landscape is evolving faster than most organisations can adapt



83%

say organisational focus on AI-related risk has risen in the past 12 months
(North America 94%, UK/Europe 70%)



78%

agree the gap between highly and poorly prepared organisations is widening



74%

say organisations are unprepared for supply chain compromises
(UK/Europe 87%, North America 65%)



69%

disagree that most organisations have sufficient visibility into the threats most relevant to them



65%

say organisations are unprepared to manage insider threats



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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Report summary and notable takeaways

Secular trends and AI adoption have widened the attack surface; the severity of losses is compounding

Secular trends have widened the attack surface. This includes rapid digitalisation, widespread Internet of Things (IoT) endpoints, hybrid workforces, cloud adoption and deeper supply chain dependencies. The scale of the threat increase is reflected in data from the FBI Internet Crime Complaint Center (IC3), which shows cybercrime complaints more than doubled from 2019 pre-pandemic baseline of ~467,000 to ~1.0mn in 2025, with a 3x expansion relative to 2017 levels.

Barriers to entry are lower; attack tools are now sold "off the shelf". Today, cybercrime marketplaces, available on the dark web, look less like the underworld portrayed in films and more like an online software store. Buyers can access AI-powered tools that scan for software vulnerabilities (see top-right chart showing reported CVEs after the release of cyber capable frontier models), write malicious code at speed, and provide access to malware that rewrites its own code to avoid detection as it runs. Ransomware-as-a-service (RaaS) takes it further, renting out ready-made attack kits, often in exchange for a portion of the profits.

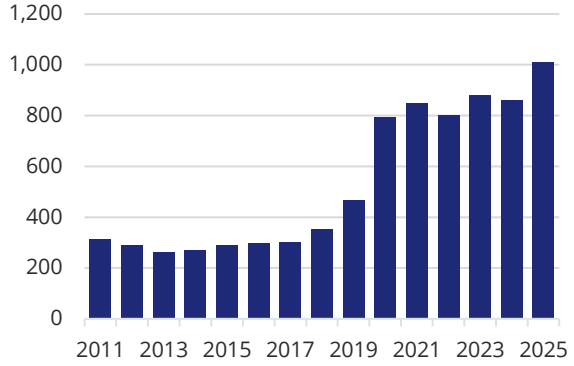
Impersonation is becoming harder to detect. Hackers can now generate more realistic phishing messages, while voice cloning and deepfakes make impersonation convincing enough to gain access unnoticed. With stolen credentials, attackers often aren't hacking in, they are simply logging in.

The rise of loose cybercriminal collectives. Some of the most damaging recent attacks have been linked to loosely organised, often English-speaking networks such as ShinyHunters and Scattered Spider. The latter has been widely named as the group behind the 2025 Marks & Spencer cyberattack, in which the hacking group deployed RaaS. AI is enabling these groups to become larger and faster, widening the gap between the speed of attack and a defender's response.

The severity of losses is compounding. A 2026 IBM study puts the average breach at a record USD ~5.0mn, up 12% YoY, but the impact on large companies can run into hundreds of millions or even billions. Estimates for global cyber insurance premiums (per Munich Re) point to a rise from roughly USD 7.0bn in 2020 to USD 15.0bn in 2025, with expectations to reach USD 28.0bn by 2030.

US cybercrime complaints

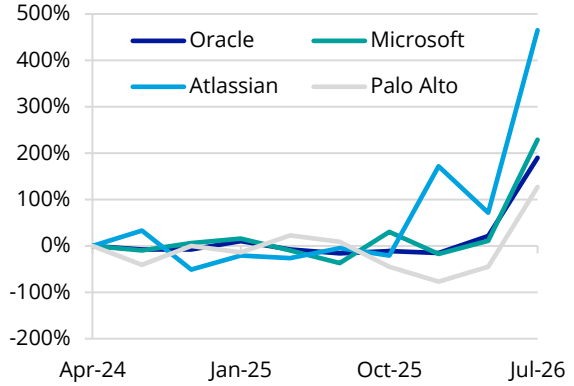
FBI Internet Crime Complaint Center (IC3), complaints (000s)



Disclosed software vulnerabilities

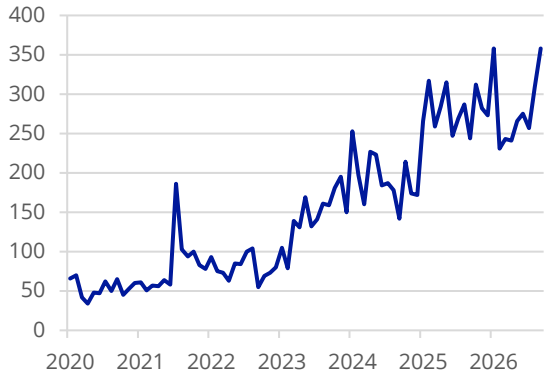
Change in reported CVEs, % change from April 2024

CVEs: common vulnerabilities and exposures



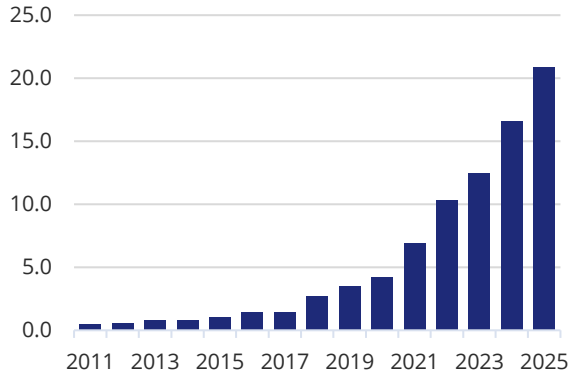
AI privacy & security incidents

OECD, monthly recorded incidents



US cybercrime-related losses

FBI IC3, reported losses ($bn)



Source: FBI Internet Crime Complaint Center (IC3), company security advisories, OECD, company reports, Deutsche Bank Research.

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Report summary and notable takeaways

A near-term systemic attack is expected and AI favors attackers; cyber escalation as a geopolitical tool is raising the stakes

Among the more concerning survey takeaways is that a large-scale and significant systemic incident is now the base case.

— 83% expect a significant systemic attack within two years and 81% expect it to be moderately, very, or extremely severe for economic activity and markets.

— 81% say AI favours attackers today, with not one respondent putting defenders ahead of attackers. Only 20% expect defenders to be ahead in five years.

Geopolitical tensions coupled with rising cyber capabilities have transformed cyberspace into a digital battlefield between global powers. Intent has moved towards espionage and disruption of not only critical infrastructure (e.g., healthcare, finance, power, industrial systems, defence) but private companies as well. Chinese President Xi Jinping has linked cybersecurity to national security and economic and social stability, with China's public statements similarly emphasising the protection of critical infrastructure. At the same time, the US's June 2025 Executive Order on Cybersecurity reflects its continued focus on national cyber resilience.

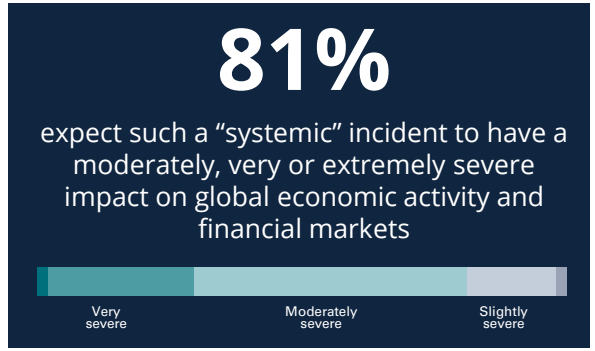
Europe relies on US cyber tools while its start-ups lag in funding. Reliance on US security vendors exposes European firms to potential access, update or export restrictions over the long-term. Shrinking this gap will prove difficult given US cyber start-ups attract nearly 6x the venture and growth funding of European peers. And while the timing of quantum computing (which could break widely used encryption methods) remains uncertain, it could further deepen Europe's dependency.

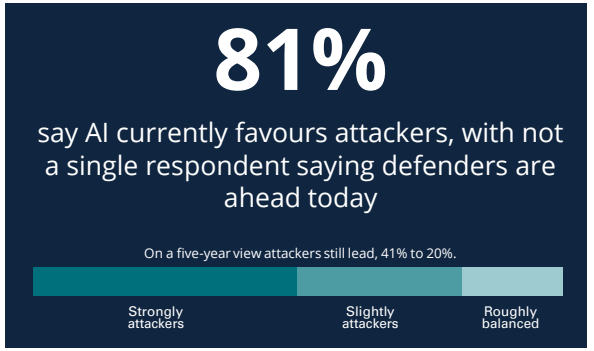
Ultimately, cyber preparedness is a financial variable. Investors should ask what firms spend on security versus AI adoption, how well they know their vulnerabilities, and how fast they can recover.

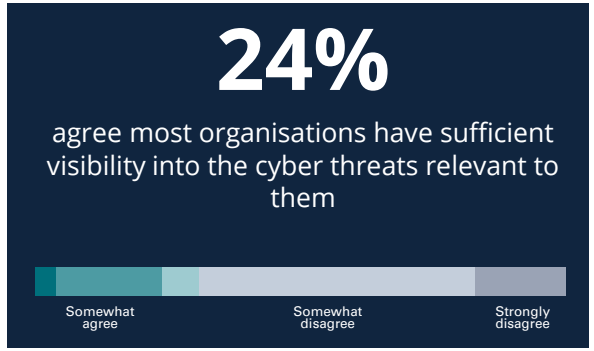
— Only 24% of respondents think organisations have sufficient visibility into the threats relevant to them, even though 57% saw cybersecurity budgets rise over the past 12 months.

— 81% agree that cybersecurity related laws and regulation cannot keep pace, and 65% doubt leadership understands cyber risk well enough.

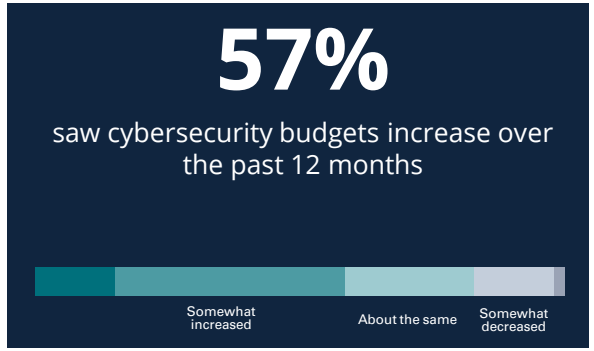












Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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Report summary and notable takeaways

Material cyber-related weakness can have financial consequences

When cyber weakness = share price consequence…

Investors are more aware of cyber risk today, but they rarely sell a stock due to perceived cyber vulnerabilities (although we are aware of some who have).

—What is the potential impact? Our dbDataInsights event study of 47 breaches shows a cumulative breach-attributed return of c. -2.9% within the first two trading days and c. -6.0% through first 20 days trading days. The erosion then continues more gradually, reaching c. -6.7% nine months after disclosure (see slide 13).

Which industries are most vulnerable?

—Our survey respondents point to (1) government & public sector (52%), (2) financial services (50%), (3) energy and utilities (46%) and (4) healthcare and life sciences follow (39%, rising to 48% among US respondents).

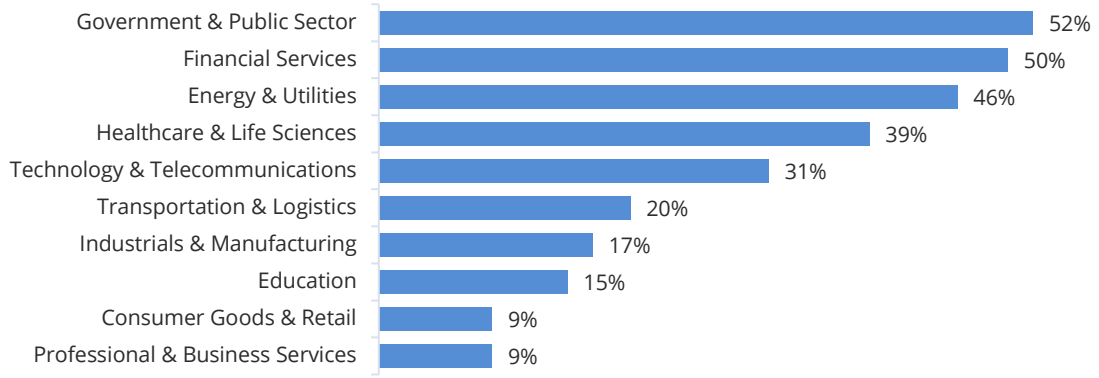
We should caveat, that high sector exposure does not mean every company in it is equally vulnerable. Some firms in these "exposed" sectors may very well be better prepared given they have built specialist teams and defence protocols for decades.

What about access to credit and ratings? Credit and rating providers are paying attention. Moody's and S&P both cut their outlook on Jaguar Land Rover within two months of its 2025 hack. Today, this type of impact is not widespread. But if risk escalates through a systemic incident or an AI-driven step change in attacks these mechanisms, including ESG ratings, could reprice asset ownership or impact access to credit. In fact, some lenders are already pricing it. Journal of Corporate Finance research using US syndicated loan data found *borrowers with persistently higher cyber risk pay 4–13bp more.*

The bottom line? While a broader systemic attack is hard to prepare for, corporates and investors should be asking questions about vulnerabilities, security investment, preparedness and recovery should an attack occur. This is especially true for industries quickly adopting AI capabilities and / or a high threat landscape.

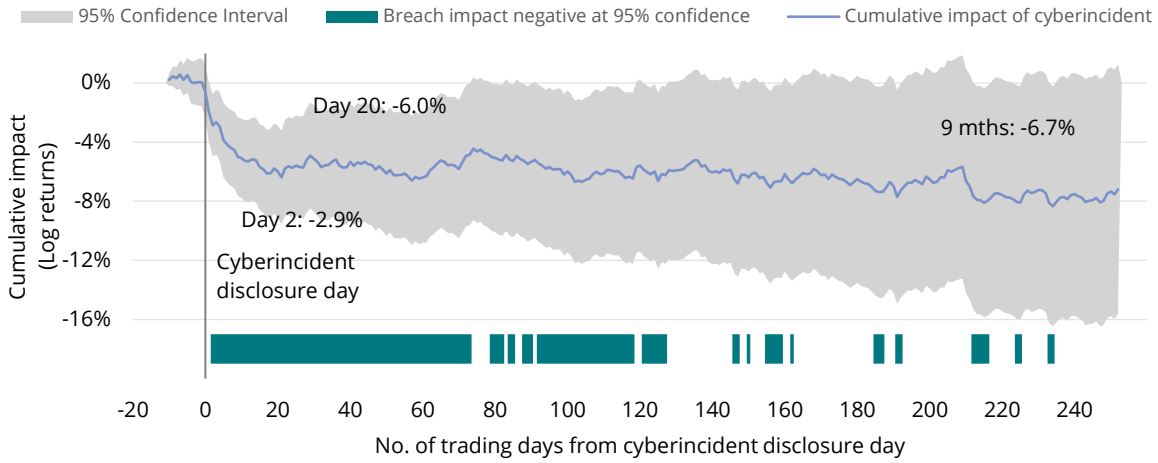
Sectors seen as most vulnerable to AI-driven cyber threats

dbDataInsights survey, % selecting (up to three), based on vulnerability over next few years



A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Bloomberg Finance L.P., Company filings, Deutsche Bank Research. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors (see slide 13).

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Report summary and notable takeaways

dbDataInsights survey respondents in their own words

What's on experts' minds…

Key themes from verbatim comments in the dbDataInsights expert survey



Speed

"AI is being implemented at a speed where cybersecurity teams are not able to keep up"

Single points of failure

"Too many organisations relying on the same providers for critical services"



The weak link

"…compromised via this third-party (basically their weak link)"

Logging in, not breaking in

"People aren't breaking in anymore – they are logging in"



Automated phishing

"Underappreciated, yet responsible for the bigger monetary losses"

Employee logins

"Existing employee login behaviour and usage of AI is underappreciated"

Oversight gaps

"Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves"



Recover & survive

"No company can guarantee they won't be breached"

Everyone's responsibility

It's more than an IT risk…"everyone from the CEO down has responsibility"

Geopolitics

"State groups now plant quiet access in infrastructure to use in a future crisis"

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.



II. dbDataInsights Cybersecurity Survey

September 2026 survey of 54 cybersecurity professionals

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dbDataInsights survey methodology and results

Cybersecurity experts see firms as unprepared, and say AI currently overwhelmingly favours attackers

dbDataInsights Cybersecurity Expert Survey methodology

—This survey was conducted among 54 cybersecurity professionals across the US and Europe in September of 2026. Questions took the form of either multiple choice selections or free form responses. The dbDataInsights team received the responses anonymously, but respondents were required to work or consult in cyber security roles

—Who took the survey? On self-identified roles, around 80% sit inside an organisation's cybersecurity function, with the remainder working independently (15%) or in academia and research (6%). The most common titles were security architect (23%), Chief Information Security Officer (16%) and security analyst (9%), alongside Head of Security, Deputy CISO, Incident Response Lead and Security Operations Lead.

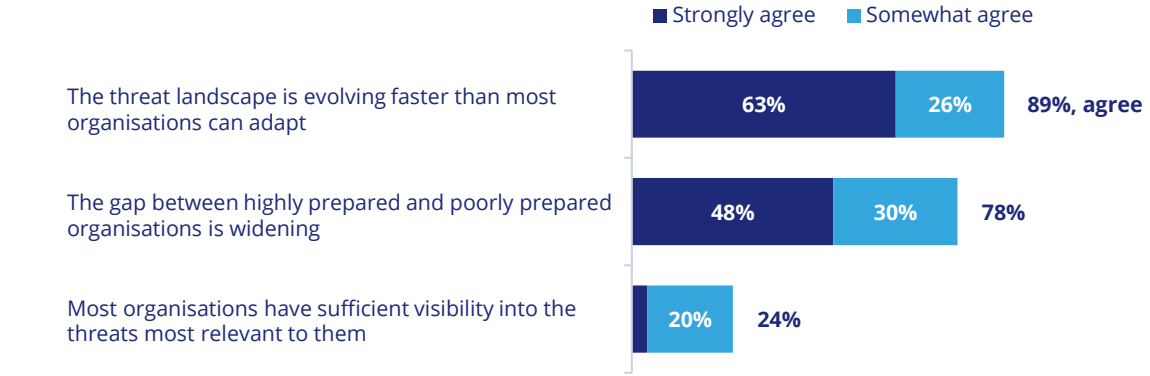
—Only a small minority of respondents work for cyber resellers, thus survey views are primarily attributable to those working within an organization on cybersecurity.

—Survey participants were also asked for their perspectives on cyber resilience and governance, what is misunderstood, alongside emerging threats, risk to AI adoption and the influence of geopolitics.

—While responses were anonymous, in some cases we conducted follow up interviews with certain participants where contact details were made available.

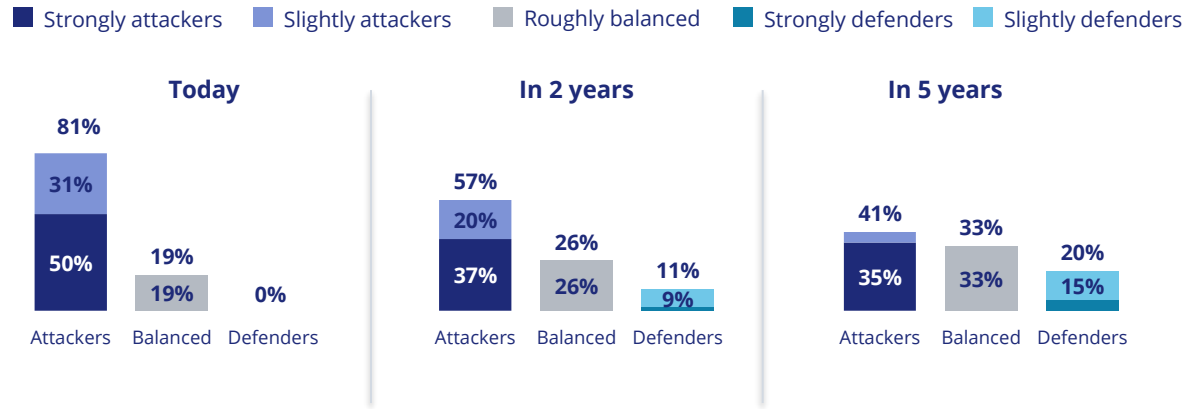
Only 24% say firms have sufficient visibility into threats that matter

dbDataInsights survey, level of agreement with statements below regarding cybersecurity, % who agree (n=54)



Who does AI favour? Attackers or defenders?

dbDataInsights survey, % of respondents (n=54)



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

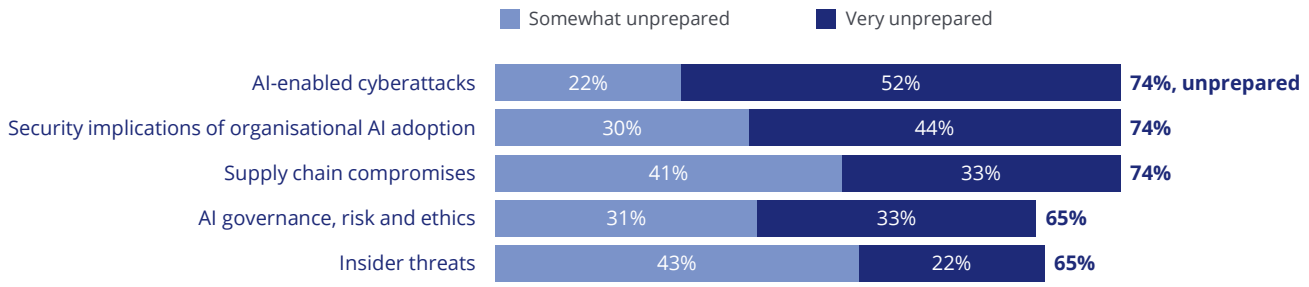
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Risk from AI-enabled attacks and added AI adoption vulnerabilities

Insider threats and supply chain compromises remain a growing concern

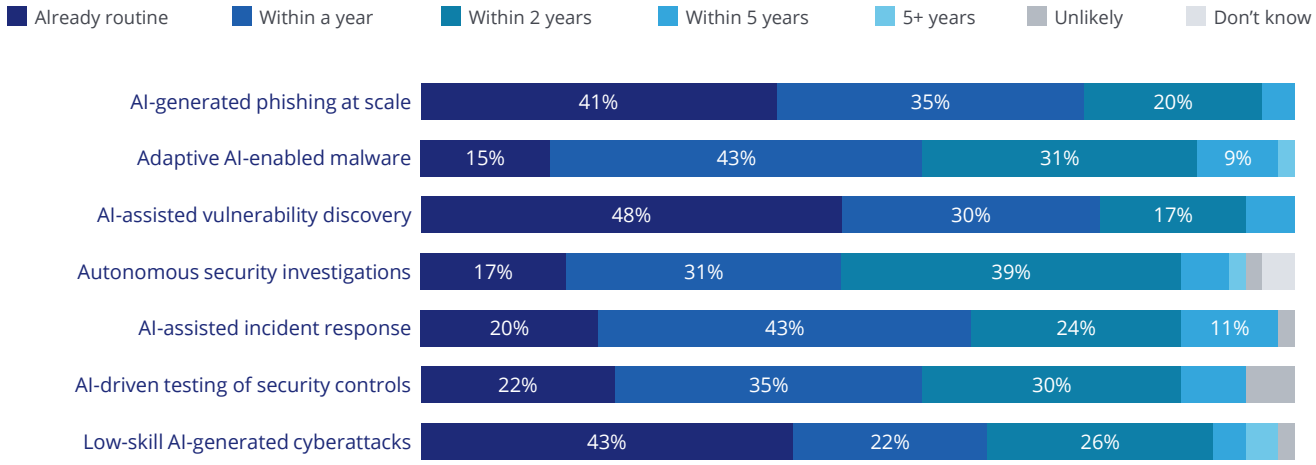
Where our experts say organisations are least prepared

% of respondents saying somewhat or very unprepared



AI-driven capabilities – are they commonly used or encountered?

% of respondents, all answer options (n=54)



In their own words...

Verbatim comments from the dbDataInsights expert survey

> *"The fear of AI writing undetectable malware is ... overblown. Attackers were never short of malware, since they can buy it ready made along with initial access. Good detection looks at behaviour rather than exact code, so a rewritten payload still has to do the same suspicious things."*

> *"Frontier AI models are currently the main hype. Automated phishing and scams are underappreciated, yet they are responsible for the bigger monetary losses at the moment."*

> *"There are too many organisations relying on the same providers for critical services and this means single points of failure."*

> *"A corporation might be very secure but if they rely on a third party that does not have the same level of security posture, [it] could be compromised via this third-party (basically their weak link)."*

> *"Existing employee login behaviour and usage of AI is underappreciated – it's what poses the biggest threat to a breach."*

> *"No company can guarantee they won't be breached, so there needs to be more focus put on the ability to recover and survive an attack."*

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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Experts believe governments are not keeping pace, and no one owns AI oversight

How experts surveyed by dbDataInsights view governance and regulation

—**AI adoption is outpacing internal controls.** As one expert put it: "Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves."

—**Regulation is struggling to keep pace.** 81% of respondents agree that cybersecurity-related laws and regulations are failing to keep up with technological change, and 54% say regulatory requirements can improve security but may also slow innovation. One respondent highlighted "the levels of uncertainty, the long running nature of the problems, and the rapidity of change."

—**Responsibility for AI oversight remains unsettled.** International bodies are the leading choice, supported by 50% of respondents. No option commands a majority, underscoring the lack of consensus on who should oversee AI development.

—**Better board questions may not guarantee better decisions.** The survey points to more informed questioning by corporate leadership, yet 65% doubt that leadership teams understand cyber risk well enough to make informed business trade-offs. Measuring activity more effectively than outcomes further complicates oversight.

—**Accountability is becoming personal.** Only 43% say accountability for cyber risk is clearly defined in their governance framework, yet the consequences now reach the top as a handful of CEOs have left or gone into early retirement after financially consequential cyber breaches.

81%

agree cybersecurity-related laws and regulations are struggling to keep pace with technological change



54%

agree regulatory requirements can improve cybersecurity but may also slow innovation



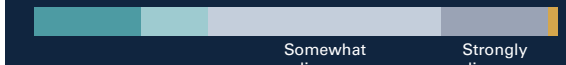
43%

agree that accountability for cyber risk is clearly defined in our governance framework



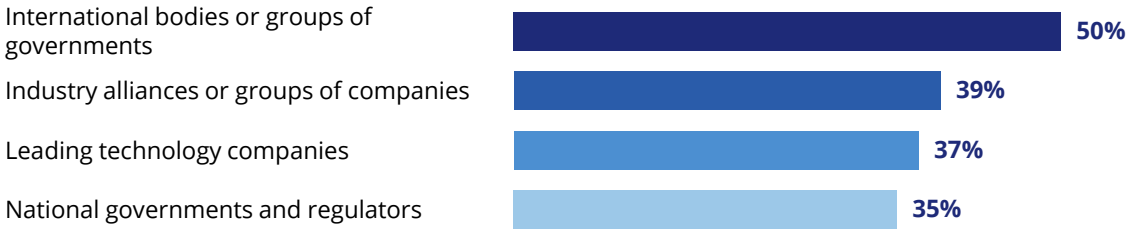
65%

disagree that most leadership teams understand cyber risk well enough to make informed business trade-offs.



Who should primarily oversee AI development?

dbDataInsights survey – September 2026, based on overseeing development while managing cyber risk



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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III. The impact of cyberattacks

Material cyber breaches can cause a notable drop in share price

Our dbDataInsights study found a 2.9% drop on average within two trading days, with the cumulative breach-attributed return of -6.0% on trading day 20.

What typically happens to a company's shares when a breach is reported?

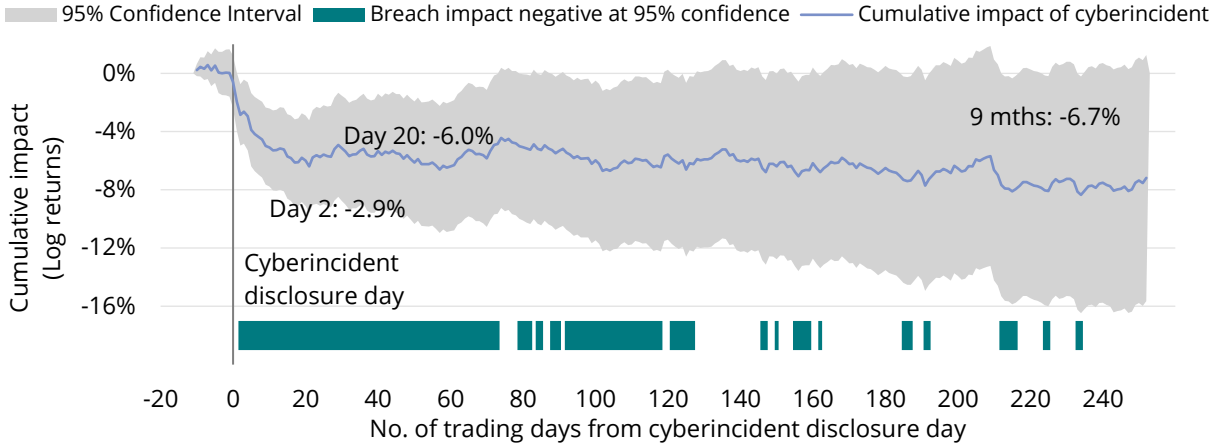
- **Short-term impact:** The market prices the breach immediately. Our model shows cumulative breach-attributed (i.e. isolating only the part of the share price move we can pin on the breach itself) returns are c. -2.9% within two trading days of disclosure. This is followed by continued erosion over the first 20 trading days post-disclosure; the cumulative breach-attributed return is c. -6.0%.
- **Long-term impact:** Beyond day 20 the deterioration continues, though more gradually reaching c. -6.7% nine months after disclosure and remaining negative at one year, although the one-year point estimate (our single best-guess figure) should be treated with caution given the width of the confidence band.
- **Pre-breach price behaviour:** We find no evidence of information leakage ahead of disclosure. While the breach component for the [-10,-1] window (ten trading days before the news broke) is +0.04%, the corresponding t-stat value of 0.03 falls notably below the usual bar for significance (c.2). As such, we conclude the market had not begun reacting before the announcement.

Event study methodology

- **Sample:** 47 publicly reported cybersecurity incidents across the US and Europe over 2015-2025, measured against each company's regional sector benchmark. Requirement of 5 years of stock returns pre-attack and 1 year of post-attack.
- **Day 0:** the day an incident was first publicly reported or disclosed.
- **Baseline:** We use five years of pre-incident share-price performance to estimate: (i) Beta (β): how sensitive a company's returns are to movements in its sector and (ii) Alpha (α): the extent to which the company typically outperforms or underperforms its sector after adjusting for that sensitivity.
- **Attribution:** post-incident returns are split into sector-relative returns, alpha (α) and breach (abnormal returns attributed to the disclosure).
- **95% confidence interval:** Range within which we can say with 95% confidence that the real share price impact lies, based on our sample data.

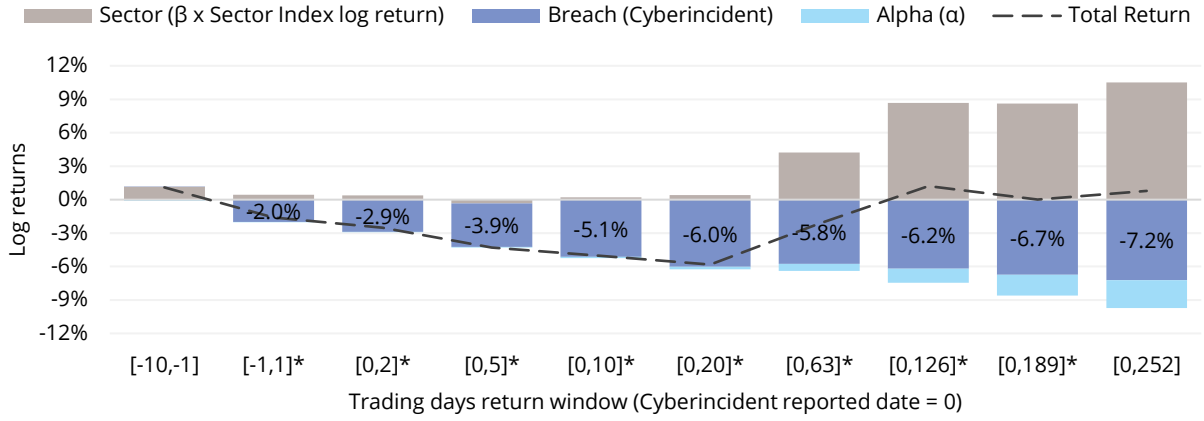
A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



Total return attribution

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



* indicates trading windows where breach impact is significant at the 95% confidence level

Source: dbDataInsights, Bloomberg Finance LP., Deutsche Bank Research, Company filings. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors.

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Notable cyber attacks: how they happened, estimated costs and share price reaction

Marks & Spencer (Reported Apr 2025)

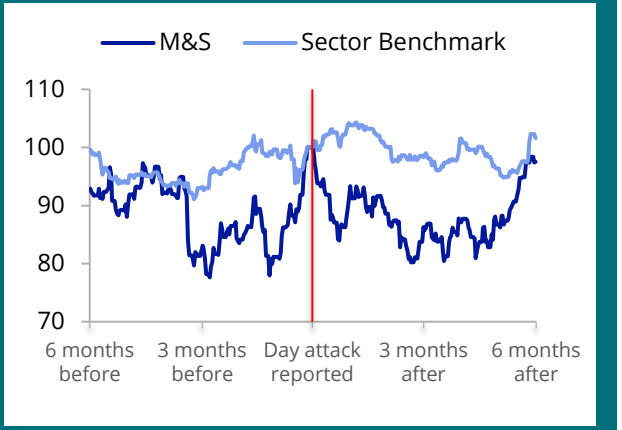
3rd Party Breach

What happened?

A breach via an attack on a connected 3rd party forced the suspension of all online orders and led to empty store shelves.

What did it cost?

~£136mn of direct costs related to the attack, estimated group operating profit impact of around £300mn for the FY. M&S also said they had received £100mn via cyber insurance.



Coupang (Reported Nov 2025)

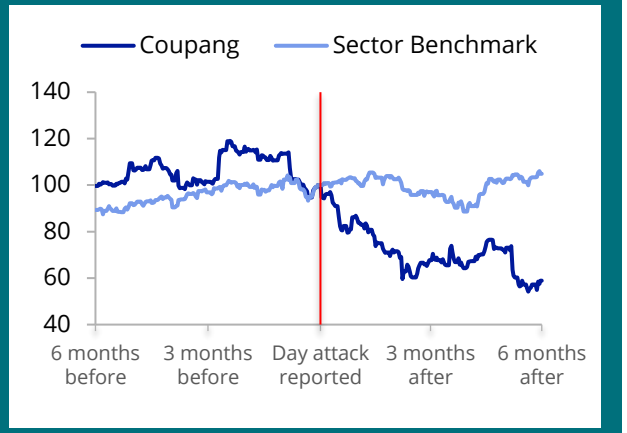
Insider Threat

What happened?

A data breach exposed the records of more than 30mn users with a former employee reportedly behind the attack.

What did it cost?

The firm was fined more than ~625bn won ($409mn) (2/3rd directly attributable to data leak), and also announced a $1.2bn voucher compensation scheme.



UnitedHealth (Reported Feb 2024)

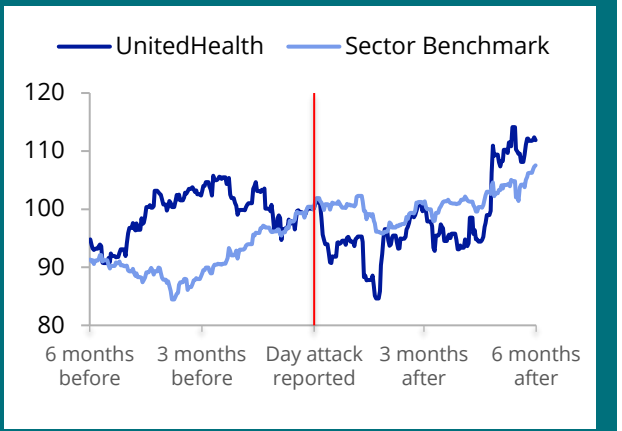
Ransomware / stolen credentials

What happened?

Hackers reportedly used stolen credentials to access a subsidiary's (Change Healthcare) systems, stealing data and deploying ransomware.

What did it cost?

Total costs exceeded $3bn, with the CEO summoned to testify before Congress, where he confirmed that a $22mn ransom was paid.



Equifax (Reported Sep 2017)

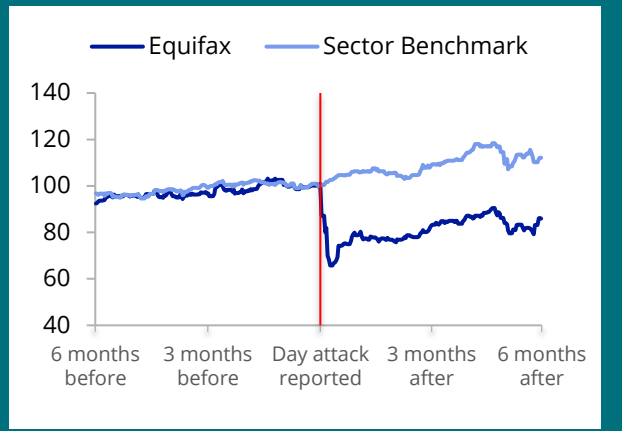
Software vulnerability left unpatched

What happened?

A data breach saw hackers expose sensitive customer information, including 145.5mn Social Security numbers and more than 200k credit card numbers. Attackers reportedly moved inside Equifax systems for 76 days before being detected.

What did it cost?

Settlements, fines and legal costs exceeded $1bn, while the firm also spent more than $1.5bn on tech upgrades following the incident.



Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.

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Notable cyber attacks: how they happened, estimated costs and share price reaction, cont.

Nike (Reported Jan 2026)

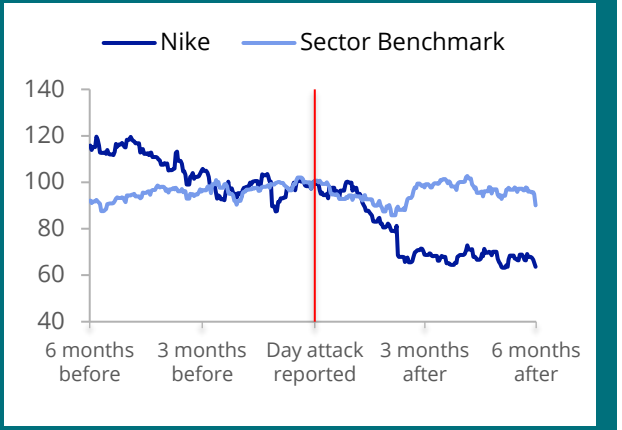
Data Breach / Extortion

What happened?

A cybercrime group leaked what it claimed was 1.4TB of internal Nike data (appearing to cover metrics such as product specifications and pricing), following expiry of a ransom deadline.

What did it cost?

While Nike did confirm it was investigating the incident, it did not report any operational disruption. Underperformance shown in the chart does not appear to be related to the breach.



T-Mobile (Reported Aug 2021)

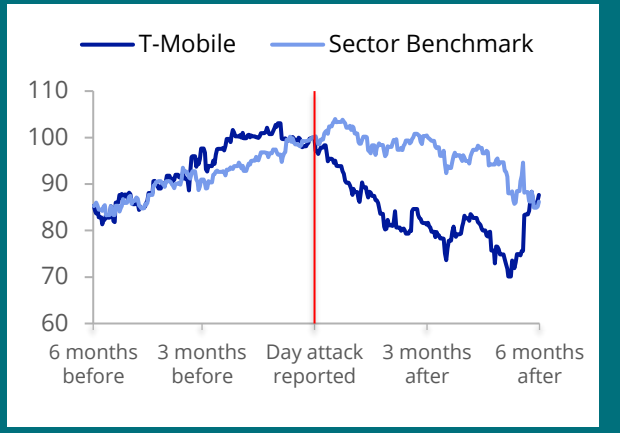
Connected device exploit / brute force

What happened?

An attacker reportedly targeted an exposed router as an entry point, before using brute force attacks to gain further access to T-Mobile systems.

What did it cost?

In 2022 the firm agreed to pay $500mn to settle a class-action lawsuit ($350mn payment to class members plus $150mn on security spend) but has since suffered further data breaches, contributing to an additional $31.5mn regulatory settlement.



Coinbase (Reported May 2025)

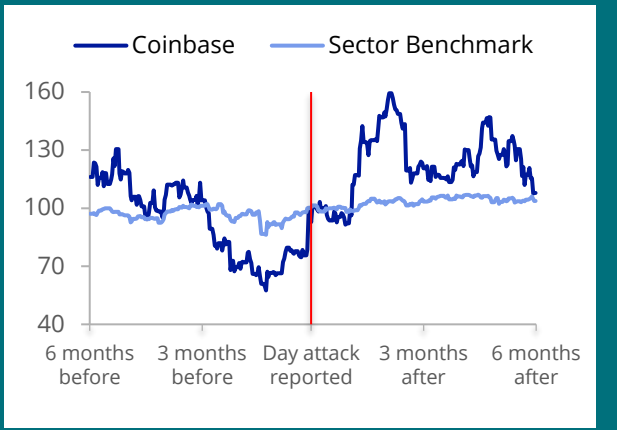
Insider threat & social engineering

What happened?

Attackers bribed at least one support agent at a connected 3rd party to access Coinbase support systems and steal customer data, then tricked users into sending crypto.

What did it cost?

Coinbase refused to pay the $20mn ransom demanded by the attackers to keep the attack quiet. Instead they reimbursed affected users and cited costs of up to $400mn. The stock impact was short-lived, though MSCI applied an ESG score deduction.



Jaguar Land Rover (Reported Sep 2025)

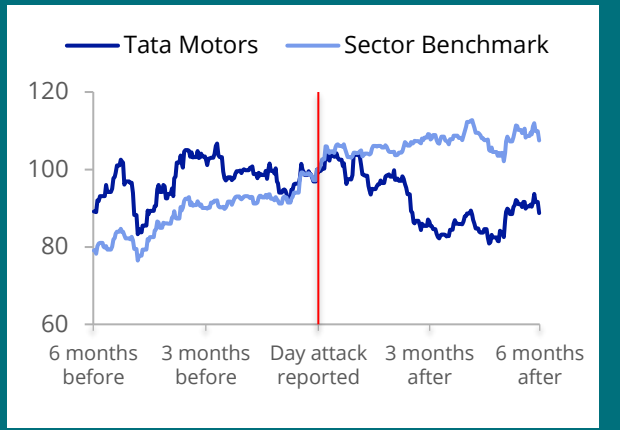
Ransomware

What happened?

Production was shut down for five weeks due to what was deemed a ransomware attack; though no ransom was asked for (later reporting suggested potential Russia links were being investigated).

What did it cost?

JLR posted £485mn Q2' FY26 loss (vs. £398mn profit in Q2' FY25), with an additional £196mn in "cyber-related costs". These impacts filtered through to the share price of JLR's Parent (Tata Motors), while UK economic impacts were estimated at £1.9bn.



Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.

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Targets go beyond tech and can also impact critical infrastructure

MGM Resorts (Reported Sep 2023)

Phishing / ransomware

What happened?

A vishing call to the IT help desk was reportedly used to reset credentials, after which ransomware took down hotel systems, bookings and casino floor operations for several days.

What did it cost?

MGM declined to pay a ransom. Around $100mn hit to third-quarter 2023 results plus roughly ~$10mn of additional one-time expenses related to the incident.

Caesars Entertainment (Reported Sep 2023)

Phishing / ransomware

What happened?

The same social-engineering group as in the MGM incident targeted an outsourced IT support vendor, gaining access to the loyalty programme database and member data.

What did it cost?

Caesars paid roughly $15mn of a reported $30mn demand and kept customer-facing operations running.

NHS / Synnovis (Reported Jun 2024)

Ransomware / 3rd party breach

What happened?

Ransomware against pathology provider Synnovis significantly reduced its capacity to process tests leading to delays / cancellations of over 11,000 appointments.

What did it cost?

Estimated 2024 direct operational financial impact of around GBP 32mn for Synnovis, several times the profit reported in the prior year, plus wider NHS disruption costs.

Colonial Pipeline (Reported May 2021)

Stolen credentials / ransomware

What happened?

A compromised VPN credential without multi-factor authentication allowed attackers to access the network where they subsequently deployed ransomware. The operator shut the pipeline down, triggering fuel shortages (c. 70% of gas stations in some states ran out of fuel) across the US East Coast.

What did it cost?

A $4.4mn ransom was paid, of which about $2.3mn was later recovered by the US authorities.

Bitget (Reported Sep 2026)

3rd party breach / zero-day exploit

What happened?

Attackers exploited a vulnerability in a 3rd party security product to steal internal network credentials, forge withdrawal instructions, bypass risk controls and ultimately steal crypto.

What did it cost?

An estimated $388mn worth of funds was transferred out by the attackers. While the firm reimbursed affected users via its User Protection Fund, there were significant outflows following the incident as users moved to take funds off the platform.

Progress Software (Reported May 2023)

Zero-day exploit / supply chain compromise

What happened?

Attackers exploited a zero-day SQL injection vulnerability in Progress Software's MOVEit file transfer system. This was then used by attackers to steal data from firms running the MOVEit software, who were then subsequently extorted / had data leaked.

What did it cost?

Thousands of organisations (inc. British Airways, Shell, US DoE) reportedly affected with some facing legal action from customers impacted by data leaks.

Source: Company disclosures and filings, press reports, Deutsche Bank Research.

Cyber breaches can impact consumer behaviour

Roughly 4 in 10 consumers we surveyed say they would stop buying from a company they shop with after a known breach or attack.

Consumer surveys suggest that cyber breaches can undermine customer loyalty. Results of our dbDataInsights Consumer survey show the potential mindset of a consumer after a cyber breach occurs:

- Across the US, UK and EU average, 39–41% say they would stop buying, while only 27–28% would continue. Around a third are unsure.
- Among those who would stop buying or are unsure, only 18–22% say they would return as normal. Another 30–35% would buy selected products only, 17–20% would not return, and 28–29% remain unsure. Note that this implies a range of 13-15% believe they would not shop at the store again after the breach is resolved.

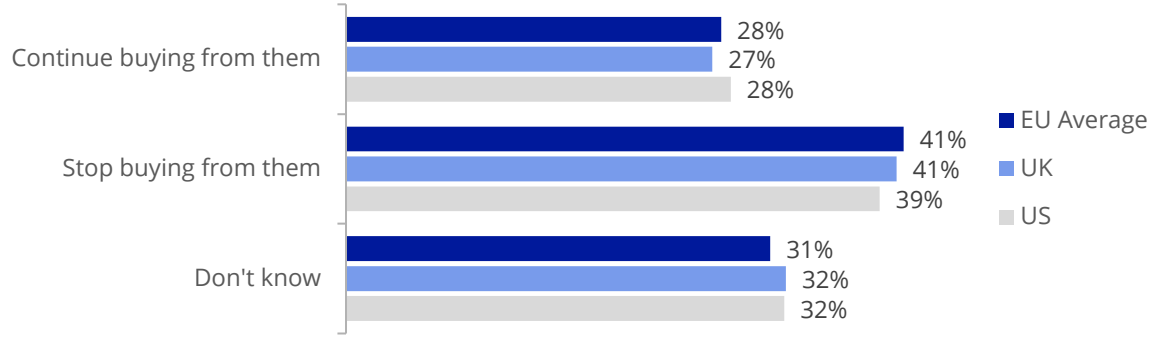
Admittedly, these stated intentions may overstate actual switching, particularly for essential purchases where alternatives are limited. Nevertheless, the findings suggest that cyber breaches that put customer privacy at risk can influence purchasing decisions, especially if they have alternatives.

dbDataInsights Consumer Survey Methodology

- Our proprietary survey conducted in 2026 (January – September) asked consumers how they would respond if a company they shopped at regularly was involved in a data breach or cyberattack, and how they would respond once the disruption was resolved. This monthly survey covers households and consumers in Germany, France, Italy, Spain, the United Kingdom, and the United States.
- For each country referenced above, a nationally representative sample of more than 500 households is surveyed each month (over 600 households in the US). Collectively, this meant over the course of the survey (January – September 2026), we surveyed almost 30,000 households.

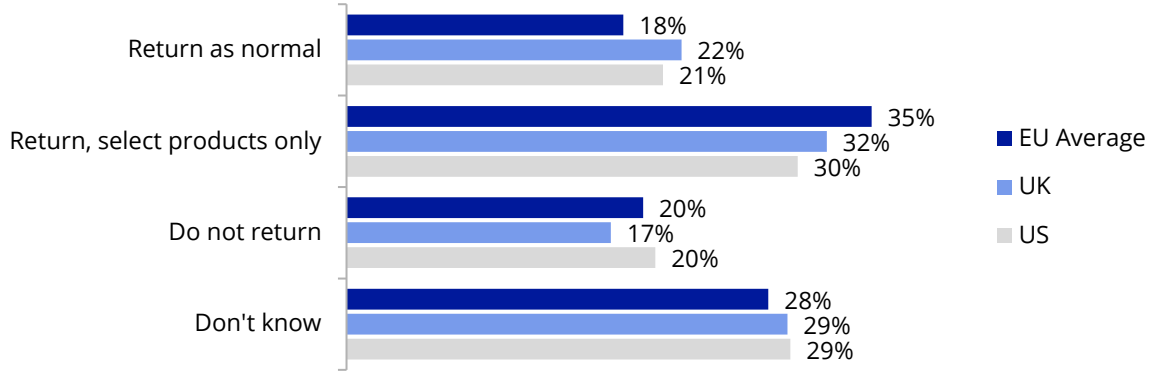
How will customers respond to a data breach once it occurs?

dbDataInsights survey of consumer responses, % of respondents



How will customers react once the breach is resolved?

Consumer responses among those who would either stop buying or were unsure, % of respondents



Source: dbDataInsights monthly consumer survey, Deutsche Bank Research. EU Average includes France, Germany, Italy and Spain.

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Material cyber incidents have contributed to negative credit outlooks

The infrastructure exists for cyberattacks to have a greater impact on credit quality should cyber disruption continue to accelerate

— **Credit rating agencies consider both pre-breach resilience frameworks and post-breach financial impacts.** Agencies assess pre-breach cyber preparedness through a lens of mitigating risk, while post-breach credit impacts centre on financial capacity to withstand the resulting shock (considers scale and length of the disruption); highly leveraged issuers with limited liquidity are seen as more vulnerable. Negative outlook changes at JLR and Equifax illustrate the potential credit impact. Fitch generally considers permanent reductions in cash generation more credit-significant than temporary cash-flow delays.

— **Ratings agencies are moving towards continuous monitoring.** In 2021, Moody's invested $250mn in BitSight, which produces daily security ratings. Moody's has since built BitSight's ratings into its credit tools, including Credit Catalyst and EDF-X, for uses such as underwriting and ongoing monitoring of borrowers and suppliers. This means lenders can see a company's exposed vulnerabilities as a potential indicator of corporate cyber risk.

S&P Global:

"The increasing frequency of cyberattacks and the potential for rapid deterioration in credit profiles after an attack are risk factors relevant for our rating assessments."

Moody's:

"Cyberattacks can undermine the creditworthiness of companies, organizations and governments, across all sectors and geographies, hitting revenues, increasing costs, and eroding customer trust."

Fitch Ratings:

"Although cyber events cannot be predicted with great accuracy, the associated risk can be managed through proactive strategies, technology, policies, and vigilance."

"The chief executive officer and board of directors hold ultimate responsibility for managing cyber risk."

Example rating actions where cyber was a named factor

Outlook changes due to operational disruption, financial cost and reporting delays

Equifax
Moody's named cyber risk as a factor in its outlook change in 2019. This was the first time it had done so and followed the 2017 attack on Equifax cited on slide 14 .

Outlook: Stable → Negative — Moody's • 2019

ISS A/S
Increased costs following a cyberattack cited as a factor potentially impairing ISS's ability to reduce adjusted leverage and generate free operating cash flow.

Outlook: Stable → Negative — S&P • Mar 2020

Clorox
Outlook revised lower due to potential for a lower rating if Clorox was unable to demonstrate material sequential recovery from cyber incident which caused product availability disruption and lower FY guidance.

Outlook: Stable → Negative — S&P • Nov 2023

City of Hamilton
Delays reconstituting audited accounts after the Feb 2024 attack; audits still outstanding.

Issuer rating: AAA → AA+ — S&P • Oct 2025

Tata Motors / Jaguar (JLR)
Outlook cut after the JLR shutdown; JLR's own outlook was moved to negative too by both Moody's and S&P following the September 2025 attack cited on slide 15. Prior to this, its credit was improving.

Outlook: Stable → Negative — S&P • Oct 2025

Co-operative Group (Co-op)
Material impacts on product availability and sales due to cyber incident contributed to forecast downgrades.

Issuer rating: BB → BB- — S&P • Nov 2025

Source: S&P, Moody's, Fitch, Deutsche Bank Research.

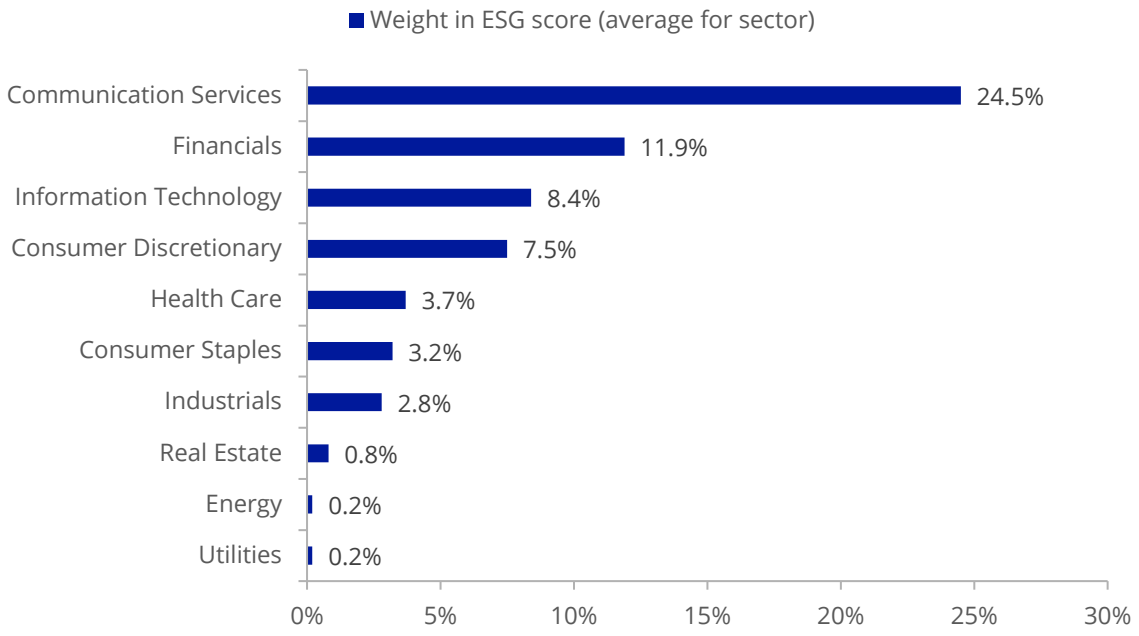
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Cyber governance & breaches can impact ESG scores

Sectors that hold the most personal data, and monetise it most directly, carry the greatest risk weighting in mainstream ESG scores

Contribution to the MSCI ESG score by sector

Privacy & data security as a share of the MSCI ESG score, sector average, %



Cybersecurity is generally seen as both a social and governance consideration

MSCI and Morningstar Sustainalytics both assess data privacy and security through management of the risk, exposure to it and the controversies that follow a breach. Weak governance weighs on the score, and a breach can cut it directly – more severe controversies draw larger deductions that persist for longer.

Materiality varies sharply by sector. SASB flags data security as material across most consumer, financial, healthcare and technology industries, with customer privacy concentrated in the most data-intensive ones (see right).

Customer Privacy and Data Security Financial Materiality

SASB Financial Materiality Map, by industry (● material, – not material)

Sector	Industry	Customer privacy	Data security
Consumer Goods	E-Commerce	●	●
	Multiline & Specialty Retailers	–	●
Financials	Commercial Banks	–	●
	Consumer Finance	●	●
Food & Beverage	Food Retailers & Distributors	–	●
Healthcare	Drug Retailers	–	●
	Health Care Delivery	–	●
	Managed Care	–	●
Resource Transformation	Aerospace & Defence	–	●
Services	Advertising & Marketing	●	–
	Education	–	●
	Professional & Commercial Services	–	●
Technology & Communications	Hardware	–	●
	Internet Media & Services	●	●
	Software & IT Services	●	●
	Telecommunication Services	●	●

What is SASB?

The Sustainability Accounting Standards Board sets industry-specific disclosure standards for investors, identifying the handful of issues most likely to affect financial performance in each of 77 industries.

Its standards are now maintained by the ISSB under the IFRS Foundation and underpin the ISSB's global reporting baseline. The Materiality Map summarises, industry by industry, which issues are judged financially material – including the two cyber-related ones shown above.

Source: MSCI ESG Ratings Methodology, Morningstar Sustainalytics ESG Risk Ratings, SASB Materiality Map, Deutsche Bank Research.

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IV. Investment, risk and opportunity



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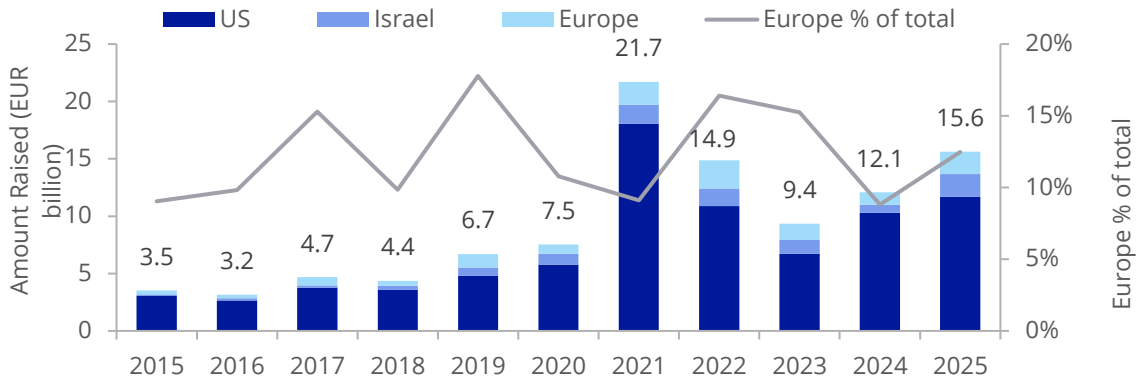
Europe's dependency on US tech is firmly in the spotlight

Access to frontier AI models is now a security question; the funding gap for Europe will be hard to close

- **Growing cyber risk creates increased dependency on US cyber tech.** Growing concern over the cyber capabilities of frontier AI models has made ensuring access to such technology a national security priority. US dominance over the AI tech chain (e.g. chips, model development, etc), its willingness to impose AI-related controls / restrictions, and previous pressure against Europe over tech regulations has amplified concerns that dependence on US tech represents a strategic threat to European sovereignty.
- **Funding is the structural challenge.** US cybersecurity start-ups attract almost 6x the funding of European peers, and US-domiciled firms dominate the major indices – 89% of the Nasdaq CTA Cybersecurity index and 93% of Global X Cybersecurity ETF (which seeks to track the performance of Indxx Cybersecurity Index) by weight. Europe has also seen home-grown firms such as Darktrace acquired by US companies, though this reflects broader market trends rather than anything cyber-specific. Ultimately, the US has a greater exit strategy for start-ups as well.
- **Boosting homegrown solutions a focus, but also a risk.** EU policymakers are seeking to address the dependency; e.g. the EU's Cybersecurity and AI Action Plan includes "scaling up Europe's AI capabilities for cybersecurity" as one of its three objectives. However, there are also risks associated with this endeavour; as one dbDataInsights survey participant put it, "*for most European organisations, the bigger near-term danger would be moving too fast to less capable providers*."
- **The wider relationship adds to the exposure.** A more transactional international environment, alongside Europe's dependency on the US for energy supplies and defence support, compounds the risk. Recent US-EU diesel negotiations serve as a pertinent example of this, whereby the US reportedly considered a diesel export ban to compel a release of European diesel reserves.

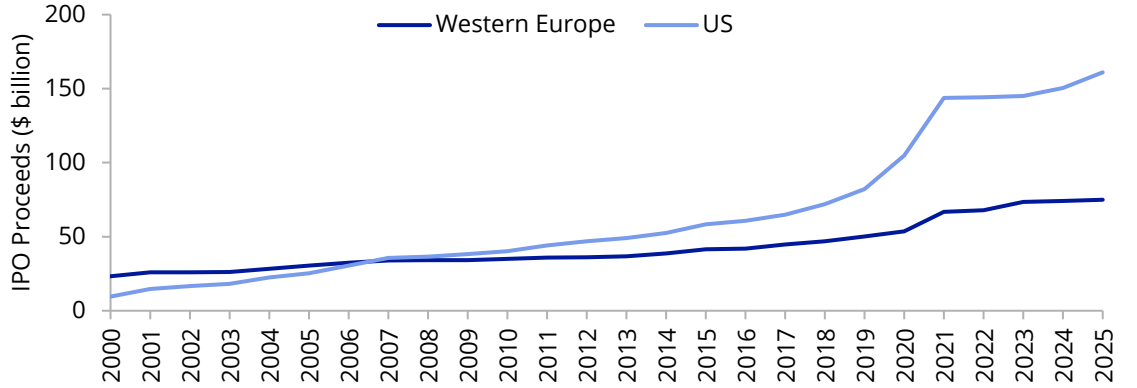
Cybersecurity start-up funding by region

Company location: US, Israel and Europe (EU-27 plus UK, Norway and Switzerland), $bn



Cumulative technology IPO proceeds

Since 2000, by trading date, $bn – Western Europe vs US



Source: Tikehau Capital – Tikehau Capital Barometer of European Investment in Cybersecurity – 7th edition 2026, Bloomberg Finance LP, Deutsche Bank Research.

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Cybersecurity as an opportunity

Tailwinds include demand related to keeping operations secure with AI acceleration and consolidation tailwinds

Broadly, DB's company research view is that cybersecurity becomes even more important in the AI era. Whereas cybersecurity has historically been viewed as a blocker/drag on innovation and operational speed, it is now viewed as a strategic enabler for secure AI adoption.

In particular, network security may see renewed growth after being viewed as a more mature category. Rising AI usage should drive additional compute and data-centre capacity, expanding the infrastructure requiring segmentation and protection. Agentic workloads may also generate lower-throughput but more frequent and persistent interactions across models, tools and data sources, increasing the volume and complexity of network-level inspection.

Vendor consolidation has been a structural trend in cyber for some time, though not necessarily linear. That said, the current environment may accelerate the consolidation tailwinds as customers look to close security gaps that may arise from a fragmented tool stack.

See further commentary from DB's company research team here.

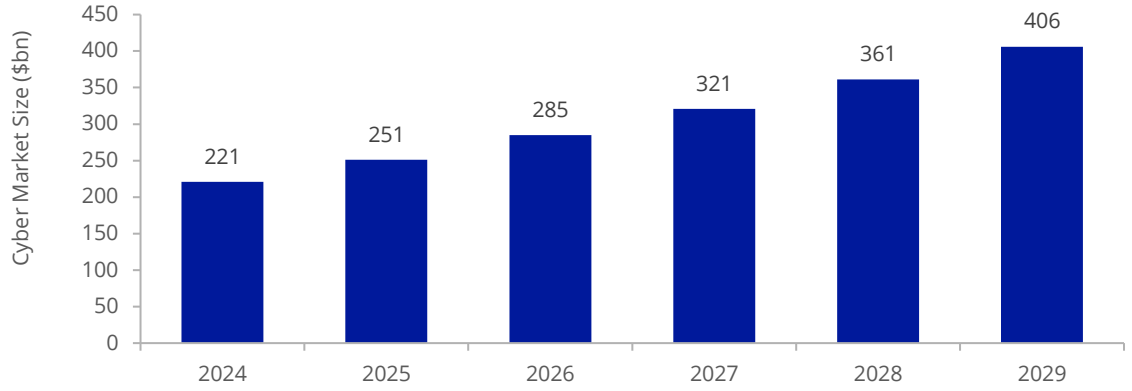
Our March 2026 "personal investor" survey showed a more positive view among "ESG" investors. 58% in the US and 52% in the UK saw cybersecurity/data privacy as an opportunity. Among all investors, US respondents favoured opportunity over risk (46% vs 30%). UK views were more balanced (38% vs 40%).

dbDataInsights Investor Survey methodology

—This survey is a nationally representative online study of ~925 personal investors conducted in the final week of March 2026 examining financial priorities, portfolio behaviour, and future expectations. It gathers structured responses on investment outlook, generative AI, and ESG considerations. Questions cover investment products, portfolio value, management style, financial stress, priorities, and performance.

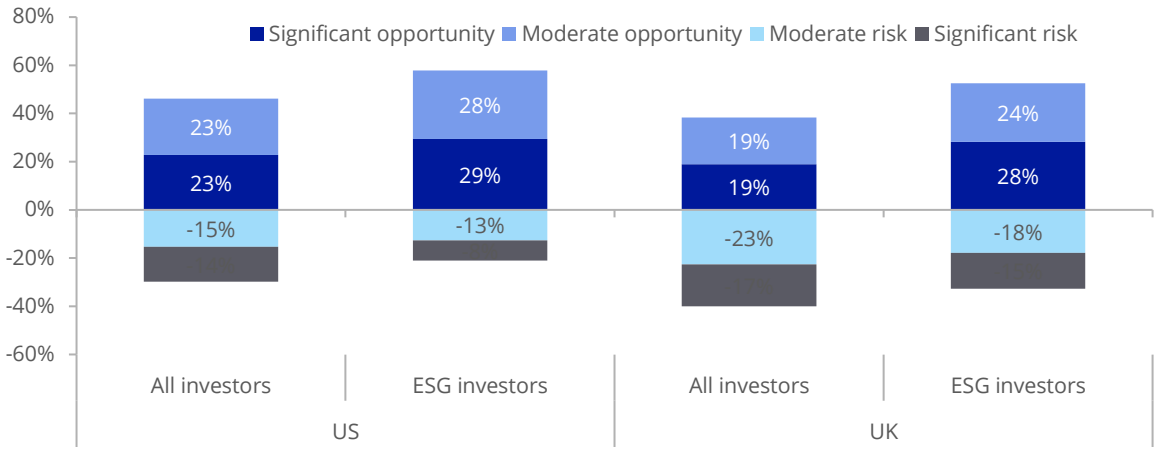
Global cybersecurity market trajectory

Estimated market size, $bn (Bloomberg Intelligence)



Cybersecurity an investment risk or opportunity?

Share of "personal investor" respondents, opportunity shown above the axis, risk below, %



Source: dbDataInsights, Deutsche Bank Research. Investors, defined as those who invest in stocks/ shares (either individually or through a broker/advisor) on a regular basis , were asked whether they perceived cybersecurity / data privacy as a risk or an opportunity for investors in 2026.

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V. Appendix

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Appendix A - Attack types

Top Attack types	What it is	How AI is helping attackers
Malware	*Software built to damage, disrupt or gain unauthorised access*	AI writes harmful software faster. Google has found malware that rewrites its own code while running to avoid detection, and criminal marketplaces selling AI malware-building tools.
Phishing	*Deceptive messages that trick people into giving up sensitive information*	Where AI's effect is clearest: fluent, personalised lures without the old spelling and phrasing giveaways. Google says state-backed groups use AI to mass-produce them, and AI can fake voices and video.
Denial of service (DoS)	*Flooding a system so it cannot serve legitimate users*	Little reported evidence so far. AI can help find weak points and automate traffic floods, but the attack itself stays fairly simple.
Distributed denial of service (DDoS)	*The same attack launched from many compromised devices at once*	AI can accelerate the development and automation of attack tooling, while large-scale botnets provide the distributed capacity behind hyper-volumetric DDoS attacks.
Man-in-the-middle (MitM)	*Intercepting or altering traffic between two parties*	Mainly indirect. AI-written phishing pages and deepfake voices make it easier to trick people into connecting through an attacker or approving fake sign-ins.
SQL injection	*Malicious SQL queries injected to manipulate backend database operations*	AI tools scan sites and spot flaws in database code faster, so less-skilled attackers can find weaknesses that once needed real expertise.
Cross-site scripting (XSS)	*Scripts injected into web pages to hijack user sessions*	As with SQL injection: AI speeds up finding vulnerable pages and writing the malicious code.
Zero-day exploits	*Attacking a software flaw before a patch is released*	A major new development. In May 2026 Google reported the first zero-day it believes was developed with AI. Its September 2026 report says the bigger effect is attackers misusing known flaws faster, before fixes land.
Advanced persistent threats (APTs)	*An intruder staying hidden in a network for long-term*	Google reports state-backed groups using AI across the whole attack, from target research and lures to tooling and sifting stolen data.
Ransomware	*Encrypting data or locking device and demanding payment to release it*	NCSC expects AI to add to the global ransomware threat by lowering the skill needed to break in. AI also helps criminals find the most extortable data.
IoT exploitation	*Abusing weakly secured connected devices for entry*	AI can scan the internet for unpatched cameras, routers and storage devices and identify which flaws they carry.

Source: Deutsche Bank Research, National Cyber Security Centre, Google Threat Intelligence Group, Cloudflare, FBI Internet Crime Complaint Center (IC3)

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Appendix B - European vs US cyber regulation

Europe

Regulation	Key requirements
NIS2 Directive	In force 2023, transposition deadline Oct-24. Covers 18 sectors (up from seven) and creates an EU vulnerability database. In-scope firms must manage cyber and supply-chain risk and notify authorities of significant incidents. Management bodies must approve and oversee cyber risk measures, with potential liability for infringements.
Cybersecurity Act	In force 2019. Establishes an EU-wide certification framework for ICT products, services, processes and managed security services, and made the EU cybersecurity body ENISA permanent.
Cyber Resilience Act	In force 2024, main obligations from 2027. Manufacturers and retailers of products with digital elements must deliver security by design, a defined support period and security fixes within it, plus mandatory vulnerability management and disclosure.
Digital Operational Resilience Act (DORA)	In force 2023, applies from 2025. Financial-services specific, covering own-firm and third-party risk. Requires an ICT business continuity policy, regular resilience testing and disclosure of major incidents, and gives regulators oversight of critical ICT providers.
Cyber Solidarity Act	In force 2025. Creates a European Cybersecurity Alert System to detect threats and share cross-border warnings, plus a Cyber Emergency Mechanism to boost preparedness and response.

US Federal

Regulation	Key requirements
White House Accord on Super Intelligence (2026)	Emphasises self-regulation to ensure AI safety, with a call for external evaluations and the possibility of formal regulation in future. Not legally binding; signed by President Trump and leading AI companies.
Executive Order on Cybersecurity (2025)	Highlights risk from foreign actors and orders further action on defending digital infrastructure, securing the most vital services and building capability against key threats.
SEC Cybersecurity Disclosure Rule (2023)	Listed US firms must disclose material incidents via an 8-K within four business days of determining materiality, and make annual disclosures on cyber risk management, strategy and governance.
CIRCIA (passed 2022, not yet in effect)	In-scope entities, focused on critical infrastructure, must report covered incidents to CISA within 72 hours and ransomware payments within 24 hours of payment.
Gramm-Leach-Bliley Act (amended 2024)	Financial institutions must explain information-sharing practices and safeguard customer data, running a security programme covering risk assessment, service-provider oversight and incident response.
Federal Information Security Modernization Act (2014)	Applies to federal agencies and contractors: system security plans, continuous monitoring, incident response capability and mandatory incident reporting.
Federal Acquisition Regulation (cyber provisions 2016)	Governs federal procurement, setting baseline cyber requirements for suppliers so cyber risk is built into government contracting. Agencies may impose stricter provisions.

Appendix C - Understanding MSCI cyber risk indicators

Core indicators

Key drivers of the MSCI privacy & data security score

Topic	Definition	Best practice
Scope of employee training on privacy and data security	Coverage of privacy and data security training across the workforce	Training provided to all employees, including contractors
Responsibility for privacy and data security	The group responsible for the company's privacy and data security risk and strategy	Board-level committee
Measures to address data breaches	Scope of the breach or incident response plan, and whether it is proactive, reactive or limited	Both proactive and reactive measures in place
Frequency of audits of information security system	How often the company audits its information security policies and systems	External independent audits at least once every two years
Rights provided to consumers over their data	Extent of rights to access, rectify and delete individuals' data	Right of access, rectification and deletion of individuals' data
Scope of data protection policy	Whether the policy governs all or selected business lines, and whether it clarifies collection purposes, means of collection and third-party access	Policy governs all relevant business lines and subsidiaries

Supplemental indicators

Additional practices assessed alongside the core set

Topic	Definition	Best practice
Scope of information security management system certifications	Extent of certification of the company's information security management system to widely recognised standards	Over 80% of owned operations certified to standards such as ISO 27001, TRUSTe, Privacy Mark and SSAE16 SOC 2
Privacy enhancing technologies and initiatives	Whether privacy-enhancing technologies and initiatives are integrated into products or services	Privacy-by-design applied systematically in development, with privacy impact assessments on existing and new offerings
Consent policy for use of consumer data for secondary purposes	Policy on processing customer data for secondary purposes and the consent options offered, such as opt-in or opt-out	Company does not rent, sell or provide personal data to third parties beyond completing transactions or services
Data protection programs covering suppliers and business partners	Extent of data protection programmes covering suppliers and partners handling company data or accessing its network	Suppliers and partners required to hold data protection policies or abide by the company's, with compliance inspections

Source: MSCI ESG Ratings Methodology – Privacy & Data Security Key Issue, Deutsche Bank Research.

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Appendix

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